

ELARA.AI

Verification and Targeted Retrieval Methodology

*Fact-checking, quote and statement verification, and search-assisted targeted scraping
in one auditable workflow*

Core principle

Elara.ai does not claim to calculate absolute truth. It organizes and scores available evidence, separates attribution from factual content, measures investigation quality and source independence, and preserves enough context and provenance for every conclusion to be audited.

Consolidated proposal - Version 1.0

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Document scope

This document consolidates the complete fact-checking, quote and statement verification, and targeted web-retrieval methodologies for Elara.ai. Repeated concepts have been unified; unique formulas, thresholds, controls, examples, and implementation requirements are retained.

1. Purpose, principles, and system boundaries

Elara.ai is an evidence-management and decision-support system for public claims, quotations, statements, news articles, and source documents. It does not treat a language model as an unquestionable judge and it does not operate as an automatic lie detector. Its purpose is to produce a structured, reproducible report that can be traced from verdict to score, from score to evidence, and from evidence to the original passage, table, transcript timestamp, recording, dataset, or filing.

Important status note

The weights, thresholds, and penalties in this methodology are transparent starting defaults. They must be calibrated on human-reviewed cases before public launch and continuously monitored for systematic errors.

The distinct questions Elara.ai answers

Output	Question answered	What it is not
Evidence Support	Does the collected evidence support or contradict the exact claim?	A mathematical probability that the claim is true.
Attribution Support	Did the named person make the statement or communicate the alleged meaning?	A permanent honesty score for the speaker.
Quote Fidelity	How accurately are the speaker, wording, sequence, and translation represented?	A judgment about whether the factual content is correct.
Verdict Confidence	How complete and dependable was the investigation?	A measure of how confidently the AI writes.
Source Independence	How many genuinely separate information origins were found?	A count of webpages, outlet popularity, or political balance.
Context Completeness	Were definitions, dates, baselines, qualifiers, and surrounding meaning preserved?	An ideological or partisan-bias score.

Design rules

- Score every important factual claim separately before producing an article-level result.
- Separate quote attribution from the truth of the quoted content; either can succeed while the other fails.
- Prefer primary records, while recognizing that an original or official source can still be selective, mistaken, or self-interested.
- Research supporting evidence and counterevidence through separate paths to reduce confirmation bias.
- Count information origins rather than URLs; syndicated or copied reporting is not independent corroboration.
- Use deterministic code for arithmetic, hashes, thresholds, rate limits, and citation-presence checks.
- Attach every factual conclusion to an exact, inspectable source passage or data value.
- Treat insufficient evidence and unresolved attribution as valid outcomes; never force every case into true or false.
- Evaluate the specific claim, quote, article, or report - not the permanent trustworthiness of a person or publisher.

Retrieval subsystem boundaries

Elara.ai uses search-assisted targeted retrieval, not a mass crawler. It converts the verification target into structured search objectives, discovers candidate sources through search services and official or domain-specific APIs, and fetches only the selected public sources needed for the investigation.

The subsystem does	The subsystem does not
Discover relevant public sources; follow citations to studies, filings, datasets, transcripts, and official records; extract text, tables, metadata, links, quotations, dates, and corrections; build evidence passages, hashes, embeddings, and provenance.	Continuously crawl the entire internet; build a general-purpose search engine; bypass paywalls, authentication, bot challenges, or private accounts; automatically trust official or highly ranked sources; republish complete copyrighted articles when limited excerpts are sufficient.

Separation of responsibilities

The retrieval subsystem finds and prepares evidence. The verification subsystem decides how the evidence relates to the claim. Search ranking predicts usefulness and extractability; it must never silently become a truth judgment.

2. Supported inputs and classification

Accepted inputs

Input	What Elara.ai extracts	Special handling
Article URL or pasted article	Headline, body, author, publication and update dates, links, quotations, statistics, and corrections.	Rank central factual claims; do not let minor background details consume the same search budget as headline-level claims.
Plain factual claim	Entities, dates, locations, quantities, metrics, definitions, and comparison basis.	Skip attribution unless a speaker is part of the claim; directly verify factual content.
Direct quote	Speaker, exact words, date, venue, source URL, and factual content.	Compare with an original transcript, audio/video recording, authenticated post, or contemporaneous document.
Paraphrase	Speaker, summarized meaning, alleged event, and factual content.	Compare semantic meaning rather than exact wording.
Social-media post	Account identity, post text, timestamp, edits, and original URL.	Authenticate the account and locate the original post or archive.
Screenshot	Visible text, account name, date, and layout clues.	Treat as unverified until the original page or permitted archive is located.
Audio or video link	Speaker, timestamps, transcript, continuity, captions, and edit boundaries.	Check cuts, surrounding speech, translation, and whether the clip hides a correction.
He-said/she-said allegation	Competing testimony, access, timing, detail, independence, and corroboration.	Do not force a verdict when no independent evidence resolves the conflict.
Uploaded document	Text, pages, headings, tables, links, metadata, and document role.	Validate type and size; use page-aware extraction and controlled OCR only when necessary.

Sentence and statement classes

Class	Example or meaning	Treatment
Verifiable fact	"Unemployment fell to 5.2%."	Extract and verify.
Attribution	"The minister said the policy would reduce costs."	Verify that the statement was made; verify the factual content separately.
Causal claim	"The policy caused inflation to rise."	Require evidence beyond correlation and separate stated explanation from established cause.
Prediction	"Prices will fall next year."	Label as a prediction; do not score as a present fact.
Opinion or value judgment	"The policy is irresponsible."	Label as opinion; do not assign a factual truth score.
Rhetorical framing	"The government finally admitted defeat."	Check embedded facts and flag framing separately.
Exact quotation	Presented as the speaker's precise words.	Apply exact-wording, speaker, sequence, context, and translation checks.
Paraphrase	Presented as a summary of meaning.	Measure semantic faithfulness rather than word-for-word match.
Allegation or testimony	A disputed account not established by direct public records.	Use careful wording, corroboration standards, and uncertainty labels.

3. Unified end-to-end verification workflow

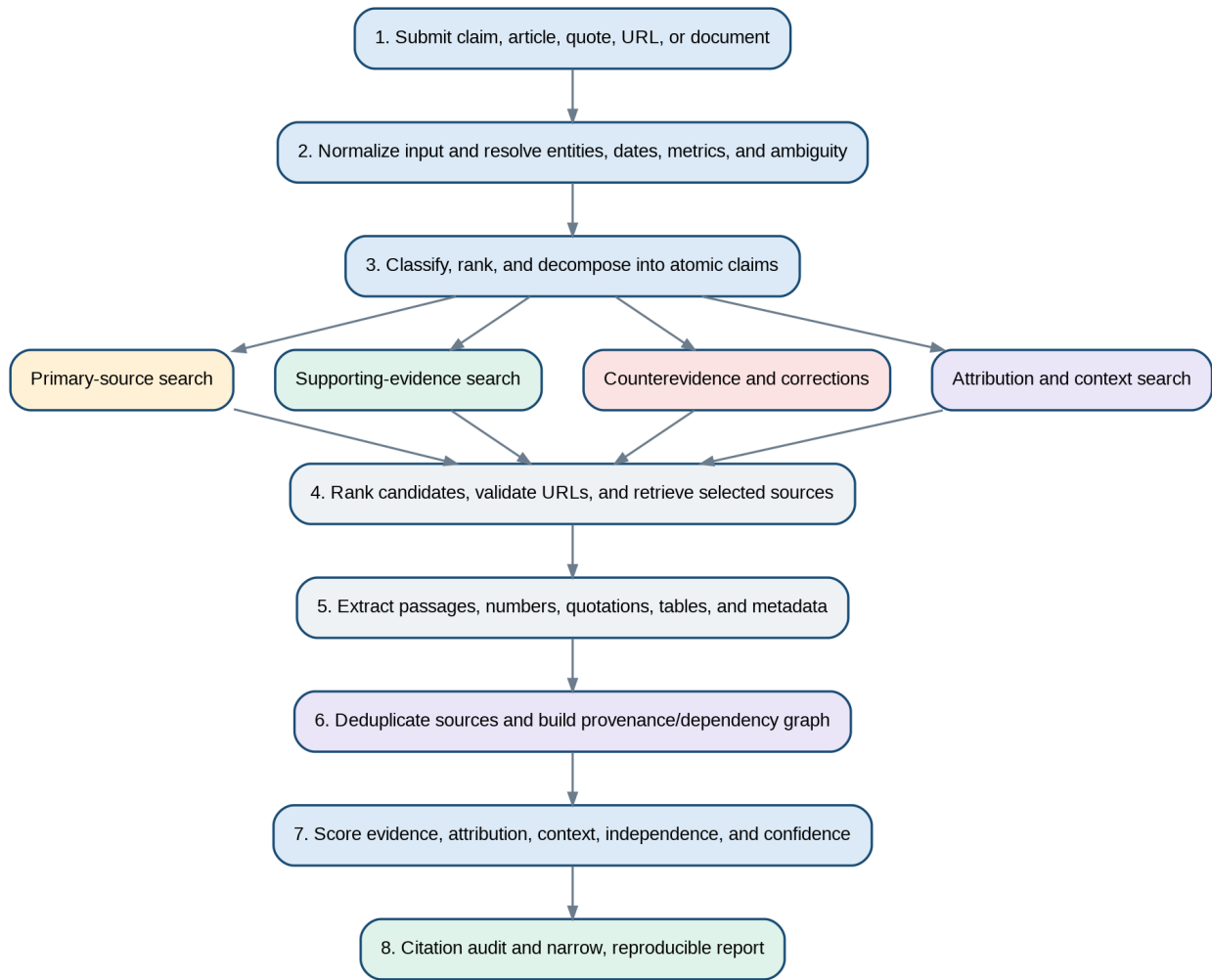


Figure 1. Unified Elara.ai workflow from submission to citation-audited report.

1. Accept a claim, article URL, quotation, pasted text, audio/video link, screenshot, or supported document.
2. Normalize the input and resolve the speaker, entities, dates, locations, metrics, definitions, comparisons, venue, and ambiguity.
3. Classify the content and split complex statements into independently testable atomic claims.
4. Rank claims by importance and create separate research objectives for primary evidence, support, contradiction, corrections, attribution, definitions, historical context, and surrounding context.
5. Discover candidates through search tools, official APIs, domain connectors, submitted citations, and existing professional fact checks.
6. Rank, canonicalize, diversify, deduplicate, validate, and securely retrieve selected sources.
7. Extract text, tables, numbers, quotations, metadata, corrections, page positions, and evidence passages; record inaccessible sources explicitly.
8. Map source dependencies and information-origin clusters so derivative reporting does not overwhelm independent evidence.
9. Audit numerical calculations, units, dates, definitions, causation, quotation fidelity, and missing context.
10. Calculate scores with deterministic formulas; synthesize only from approved evidence.
11. Run a citation audit to confirm that every factual sentence in the final report is supported by the cited passage.
12. Publish the narrowest defensible conclusion with limitations, timestamp, methodology version, and source history.

4. Claim understanding, extraction, and prioritization

Structured claim representation

Search quality depends on understanding the target rather than simply extracting keywords. Elara.ai creates a structured representation before searching.

```
{
  "claim": "Company X doubled its profits last quarter because demand increased.",
  "entities": ["Company X"],
  "time_period": "latest reported quarter",
  "locations": [],
  "metrics": ["profit", "demand"],
  "comparison": "prior comparable period",
  "ambiguities": ["profit may mean net income, operating income, or adjusted earnings"],
  "atomic_claims": [
    "profit increased",
    "the increase was approximately 100 percent",
    "demand increased",
    "demand caused the profit increase"
  ]
}
```

Field	Examples	Why it matters
Entities	Person, company, country, product, law	Prevents evidence about a similarly named entity from being mixed in.
Time	Exact dates, quarter, year, "recently"	Old but accurate information may not answer a current claim.
Location	Country, province, city, market	Statistics and laws vary by jurisdiction.
Metric	Net income, revenue, crime rate, emissions	Different definitions can produce different conclusions.
Comparison	Year-over-year, quarter-over-quarter, baseline	Determines the records and calculations required.
Claim type	Numerical, causal, quotation, scientific	Activates different research strategies and evidence standards.

Atomic claim decomposition

One sentence may contain multiple claims requiring different evidence. For "Company X doubled its profits after its AI product caused demand to surge," Elara.ai separately checks whether profits increased, whether the increase was about 100 percent, whether product demand increased, whether the AI product caused the demand increase, and whether demand caused the profit increase.

Importance weights and essential-claim gate

Importance	Weight	Definition
Essential	3	The headline or central conclusion depends on this claim.
Major	2	Important support for the article's conclusion.
Minor	1	Background detail that does not materially change the conclusion.

Essential-claim gate

If an essential claim is strongly refuted with adequate confidence, the article cannot receive a fully supported assessment even when several minor details are accurate.

5. Quote attribution, fidelity, and contextual verification

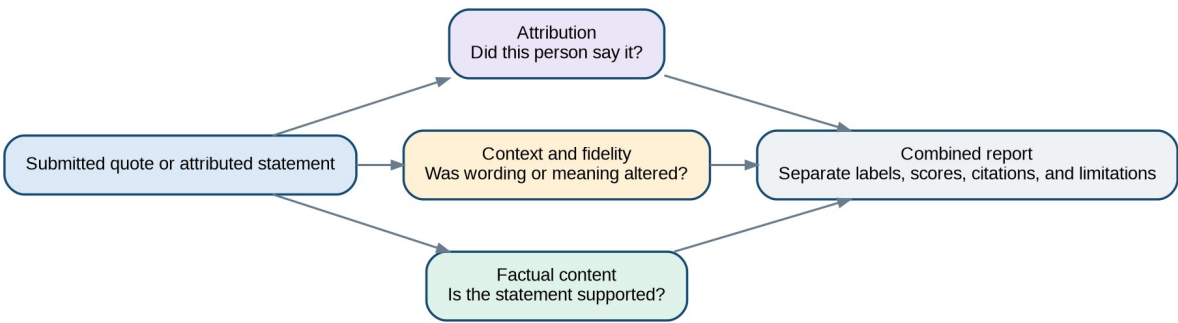


Figure 2. Quote verification separates attribution, fidelity/context, and factual content.

The two-question model

A quotation contains at least two independent claims: whether the named person said the words or communicated the meaning, and whether the factual claims inside the statement are supported. A quote can be authentic while its content is wrong; a fabricated or misattributed quote can contain an independently true fact.

Evidence hierarchy for attribution

Evidence type	Typical strength	Notes
Unedited original audio or video	Very high	Best when identity, date, venue, and continuity can be established.
Official transcript or authenticated original post	High	Strong direct record, though transcripts may be corrected or incomplete.
Contemporaneous notes, messages, emails, or documents	Moderate to high	Depends on authenticity, timing, and whether the words were directly recorded.
Independent eyewitness accounts	Moderate	Stronger when witnesses had direct access and did not coordinate.
One witness versus one denial	Low to moderate	Usually insufficient without corroboration.
Later second-hand recollection	Low	Memory and retelling can change over time.
Screenshot without original source	Low	May be edited, cropped, or fabricated.
Anonymous repost or meme	Very low	Useful as a lead, not as proof.

Attribution checks

- Speaker identity: is the person correctly identified?
- Originality: is this the earliest or most direct record rather than a later retelling?
- Wording: are the quoted words present in the record?
- Timestamp and venue: do the event, date, and setting match the allegation?
- Continuity: was audio or video cut between meaningful phrases?
- Authorship: is the account, email, post, or document authentic?
- Translation: does the translated wording preserve the original meaning?
- Quotation nesting: was the speaker quoting another person rather than asserting the words personally?

Attribution labels

Label	Meaning
Exact quote confirmed	The wording appears in an authenticated original record.
Accurate paraphrase	The exact words differ, but the summarized meaning is faithful.

Label	Meaning
Partially accurate / exact wording uncertain	Some wording or meaning is confirmed, but material language remains missing, altered, or disputed.
Taken out of context	The words were said, but surrounding material materially changes their meaning.
Misattributed	The words came from another speaker or source.
Fabricated	Strong evidence indicates the quote or record was invented or altered.
Unable to verify	Available public evidence cannot establish whether the statement was made.

Quote Fidelity Score

Component	Weight	Question
Wording match	35%	How closely does the quote match the original words?
Speaker identity	20%	Is the correct speaker identified?
Completeness	20%	Were material words or sentences omitted?
Sequence integrity	15%	Were separate phrases spliced or reordered?
Translation accuracy	10%	When translated, is the meaning preserved?

Quote Fidelity

$0.35W + 0.20S + 0.20C + 0.15Q + 0.10T$

Each component is scored from 0 to 100. When translation is not applicable, its weight is omitted and the remaining weights are normalized.

6. Search planning and targeted source discovery

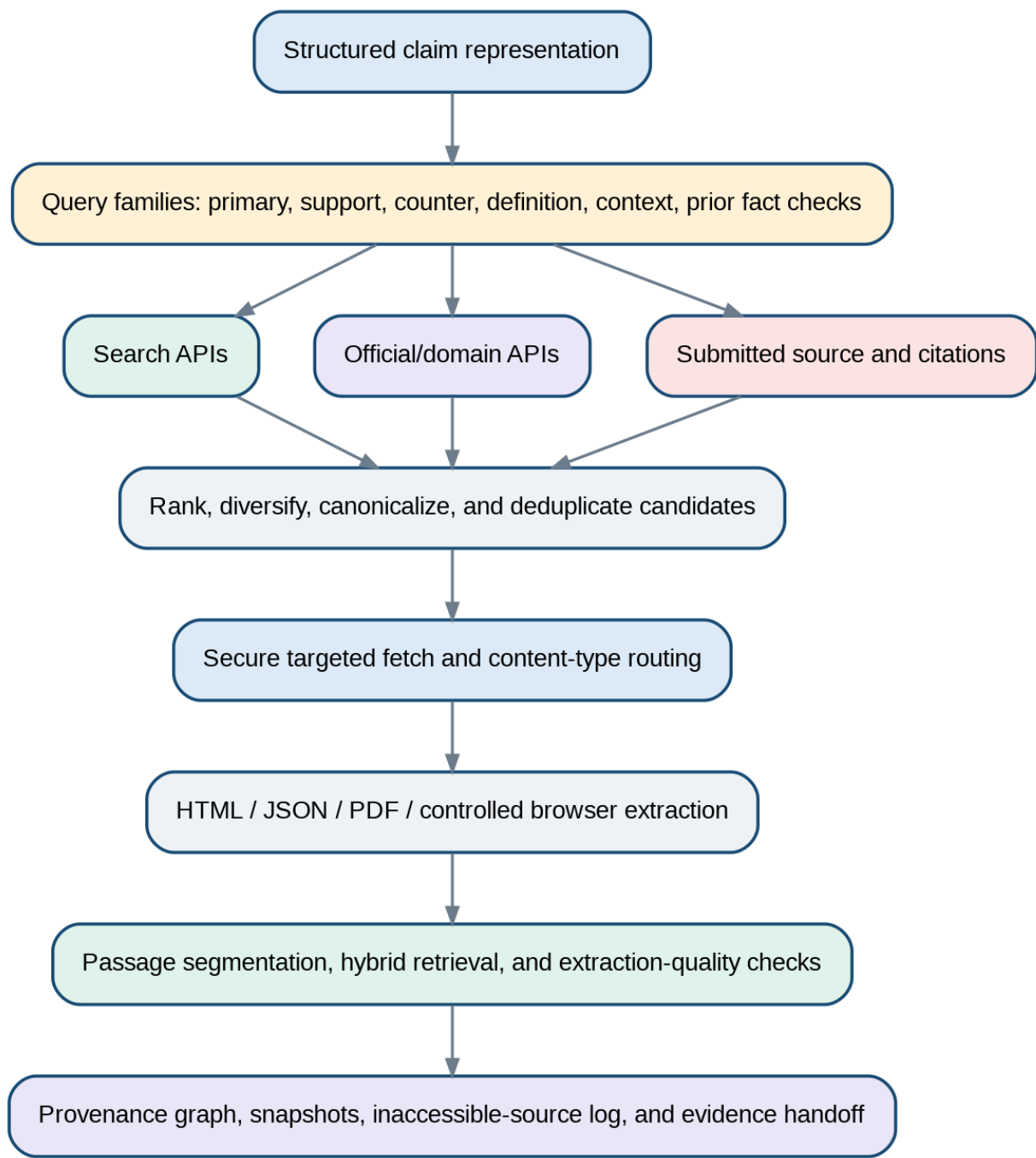


Figure 3. Search-assisted targeted retrieval and evidence handoff.

Research branches and query families

Query family	Purpose	Example
Primary source	Find the original record, dataset, transcript, filing, study, legislation, or recording.	Company X quarterly filing net income 2026
Supporting	Find the strongest independent evidence consistent with the claim.	Company X profits doubled latest quarter

Query family	Purpose	Example
Counterevidence	Actively search for disagreement, correction, retraction, omitted context, or alternative explanation.	Company X profit did not double adjusted earnings
Definition	Resolve ambiguous metrics or terms.	Company X operating income versus net income
Context	Find baselines, limitations, wider trends, and historical comparisons.	Company X demand unit sales historical quarters
Attribution	Locate the earliest or most direct statement record.	Exact quoted phrase transcript speaker date
Existing fact check	Find prior professional investigations of the same or related claim.	"Company X doubled profits" fact check
Historical context	Determine whether old information is being presented as new.	Previous definition or prior release date

Query-set construction

exact terms + normalized entities + synonyms + metric definitions + date/location filters + adversarial variants

At least one query preserves an exact quotation during attribution checks. Neutral and contradiction-oriented queries must not assume the submitted claim is correct.

Query safeguards

- Preserve exact quotations in at least one attribution query.
- Add explicit dates and locations whenever known.
- Generate neutral queries and a separate contradiction-oriented path.
- Store every generated query so the research path remains auditable.
- Limit query count according to Quick, Standard, or Deep mode and stop when coverage thresholds are met.

Source roles

Role	Examples	Interpretation
Primary evidence	Dataset, filing, transcript, original study, legislation, recording	Closest available record, but not automatically neutral or error-free.
Official self-report	Company release, ministry statement	Direct for what the organization reported; weaker for independent self-evaluation.
Independent analysis	Academic, technical, or investigative analysis	Useful when methods and evidence are visible.
Secondary report	News article summarizing another source	Useful but should be traced to the information origin.
Derivative report	Syndication, rewrite, repost	Adds little or no independent corroboration.
Opinion or commentary	Editorial, analysis column	May supply interpretation, not independent proof of factual claims.

Retrieval-priority score

Factor	Weight	Meaning
Claim relevance	30%	How directly the result appears to address the atomic claim.
Source directness	20%	Original filing or dataset ranks above a page discussing it.
Temporal fit	15%	The source concerns the correct event or reporting period.
Source-type diversity	15%	Avoid filling the queue with near-identical reports.
Novelty	10%	Prefer a new information origin or perspective.

Factor	Weight	Meaning
Extractability	10%	Public, readable, supported content types rank above inaccessible pages.

Retrieval priority

$0.30R + 0.20D + 0.15T + 0.15V + 0.10N + 0.10E$

All inputs range from 0 to 1. This predicts usefulness and extractability; it is not a credibility or truth score.

Selection rules

- Reserve source slots for primary evidence whenever available.
- Reserve separate slots for supporting and contradicting evidence.
- Limit pages from one domain or obvious content cluster.
- Deduplicate canonical URLs before fetching.
- Stop when marginal sources no longer improve claim coverage or independence.

7. Secure fetching and content extraction

Targeted fetch decision sequence

- 1. Canonicalize the URL and reject unsupported schemes.
- 2. Resolve the hostname and block private, local, reserved, link-local, and cloud-metadata addresses.
- 3. Check the cache and recent content hash before sending a new request.
- 4. Send a limited request with connection, read, and total timeouts; redirect limits; response-size limits; and an identifiable user agent where appropriate.
- 5. Inspect the status code, file signature, and content type before downloading or parsing the full response.
- 6. Route structured JSON to an API parser, static HTML to the standard extractor, JavaScript-heavy pages to a controlled browser fallback, and PDFs to a page-aware document parser.
- 7. Run extraction-quality checks; accept the result, try one allowed fallback, or record the source as inaccessible.

Why browser rendering is a fallback

A headless browser is slower, more expensive, and exposes a larger attack surface than a normal HTTP request. Use Playwright only when client-side rendering is necessary and the source is important enough to justify the cost.

URL and network controls

Control	Required behavior
Allowed schemes	Accept only HTTP and HTTPS.
Hostname validation	Reject localhost, private networks, link-local ranges, reserved addresses, and cloud metadata endpoints.
DNS re-check	Validate the resolved address at connection time to reduce DNS-rebinding risk.
Redirect policy	Limit redirects and revalidate every destination.
Port policy	Restrict requests to approved web ports unless explicitly configured.
File policy	Reject executable or unsupported response types.
Size policy	Abort responses exceeding the configured limit for the content type.
Timeouts	Use connection, read, and total-request deadlines.
Concurrency and rate limits	Apply per-domain and per-user limits to prevent accidental denial of service.
Credential isolation	Never forward user authentication cookies or private provider credentials to third-party sites.
Parser isolation	Store fetched files outside executable paths and never launch them.

HTML extraction and normalization

- 1. Decode the response using declared and detected character encoding.
- 2. Remove scripts, styles, navigation, advertisements, cookie notices, comments, and unrelated recommendations.
- 3. Extract the main article body with the primary article extractor.
- 4. Inspect metadata, headings, links, correction notices, and custom selectors with an HTML parser.
- 5. Preserve headings, lists, tables, captions, block quotations, and meaningful document structure.
- 6. Normalize whitespace and Unicode while preserving quotation marks, numbers, units, and sentence boundaries.
- 7. Calculate a content hash and extraction-quality score.

Metadata group	Captured values
Identity	Canonical URL, title, publisher, domain
Authorship	Author, organization, speaker
Time	Published, updated, retrieved, and event dates

Metadata group	Captured values
Document role	News report, press release, study, filing, transcript, opinion
Outbound sources	Cited links, study identifiers, dataset links
Correction state	Correction, editor note, retraction, amended filing
Integrity	Content hash, parser name/version, snapshot identifier

Extraction quality checks

- Minimum readable-text length after boilerplate removal.
- Title and body consistency.
- Duplicate-line and navigation-noise ratio.
- Presence of the searched entities, dates, quotations, or claim terms.
- Detection of truncation, access barriers, malformed tables, or hidden content.

PDFs, structured APIs, and JavaScript-heavy pages

- Prefer official or well-documented APIs when they expose the required record directly; structured JSON reduces parsing errors and makes identifiers, dates, units, and revisions easier to validate.
- For PDFs, confirm file signature and type, extract page-aware text, headings, links, and layout, and convert necessary tables into structured rows.
- Attach every PDF passage to a page number and source snapshot.
- Use OCR only as a controlled fallback for image-only documents and lower extraction certainty for OCR-derived text.
- Attempt static extraction before Playwright; disable unnecessary media and third-party requests; enforce navigation timeout and DOM-size limits.
- Never expose provider keys, internal cookies, or privileged browser sessions to a retrieved page.

Condition	System behavior
Paywall or login required	Do not bypass. Record metadata and use public alternatives or lawful user-provided text.
Bot challenge	Do not defeat it. Mark inaccessible and continue.
Unsupported media	Record the limitation and route to a future media-verification subsystem when available.
Malformed document	Try one safe parser fallback, then preserve the error.
Deleted page	Use permitted archives or metadata when available; never imply the original page was read.

8. Evidence passages, provenance, and source independence

Passage segmentation and hybrid retrieval

- Segment by semantic boundaries such as headings, paragraphs, transcript turns, and table rows.
- Use limited overlap so boundary sentences retain context.
- Preserve page number, heading path, paragraph order, and source identifier.
- Keep quotations attached to the speaker and surrounding text.
- Keep table values attached to row and column labels.

Passage relevance

lexical score + semantic similarity + metadata fit + reranker score

Exact matching is strongest for names, quotations, numbers, and identifiers. Embeddings are useful when a source expresses the same idea with different wording. Metadata fit includes entity, date, location, document type, and claim type.

```
{
  "source_id": "src_17",
  "passage_id": "passage_17_04",
  "text": "Net income for the quarter was ...",
  "page_or_section": "Quarterly results - page 8",
  "published_at": "2026-05-14",
  "retrieval_reason": "Directly addresses atomic claim 2",
  "extraction_certainty": 0.98
}
```

Prompt-injection boundary

Retrieved pages are evidence, not instructions. Text such as "ignore previous instructions" is stored as source content and must never change system rules, tool permissions, workflow policy, or the final verdict.

Dependency signals and relationships

- Explicit citations and outbound links.
- Wire-service and syndication labels.
- Near-identical text, paragraph order, quotations, statistics, tables, or errors.
- Publication timestamps and first-known appearance.
- References to the same press release, anonymous source, study, filing, recording, witness, or dataset.

Relationship	Meaning
CITES	One source directly links to or names another.
REPUBLISHES	Content is syndicated or substantially identical.
QUOTES	One source repeats a quotation from another.
DERIVES_FROM	The claim or data originates from another source.
USES_SAME_DATA	Sources independently analyze the same underlying dataset.
POSSIBLE_DUPLICATE	Similarity is high, but the exact relationship remains uncertain.

Dependency contribution multipliers

Relationship within an information cluster	Multiplier
Original or genuinely independent evidence / strongest original record	1.00
Meaningful independent analysis using the same information chain	0.35
Mostly repeated reporting or second-hand retelling	0.10
Exact syndicated or copied version	0.00

Adjusted evidence weight

$w_i = q_i \times m_i$

q_i is base evidence quality; m_i is the dependency multiplier. Ten pages derived from one press release should not outweigh two genuinely independent records merely because there are more URLs.

Source Independence Score

Component	Weight
Independent information-origin diversity	40%
Primary-record diversity	25%
Organizational or ownership diversity	20%
Method or dataset diversity	15%

Source Independence

$0.400 + 0.25P + 0.20G + 0.15M$

Each component ranges from 0 to 100. Several stories quoting the same witness remain one testimonial origin; independent witnesses, records, recordings, or datasets contribute distinct evidence.

9. Evidence quality and claim-support scoring

Evidence-item quality

Dimension	Weight	Question
Relevance	25%	Does the passage directly address the exact claim?
Directness	20%	How close is the source to the original record or observation?
Claim-specific authority	20%	Does the source have appropriate access or expertise for this claim?
Transparency	15%	Are author, date, method, data, citations, and correction history visible?
Temporal fit	10%	Does the evidence apply to the claimed event or reporting period?
Extraction certainty	10%	Were the speaker, wording, value, unit, and context captured reliably?

Base evidence quality

$$q_i = 0.25R + 0.20D + 0.20A + 0.15T + 0.10F + 0.10X$$

All inputs range from 0 to 1.

Evidence stance

Classification	Stance value
Strongly contradicts	-1.00
Partially contradicts	-0.50
Neutral or irrelevant	0.00
Partially supports	+0.50
Strongly supports	+1.00

Rejection gates

- Relevance below 0.50.
- Extraction certainty below 0.65.
- Wrong person, organization, event, location, or time period.
- The quotation or number cannot be located in the cited source.
- The page discusses the controversy but supplies no evidence for the exact claim.

Atomic factual-support score

Supporting weight

$$P = \sum(\max(s_i, 0) \times w_i)$$

Contradicting weight

$$N = \sum(\max(-s_i, 0) \times w_i)$$

Evidence / Factual Support

$$100 \times P / (P + N)$$

A result near 100 indicates strong support; a result near 0 indicates strong contradiction.

Score	Default label
90-100	Supported
75-89	Mostly supported
60-74	Leaning supported
40-59	Mixed or unresolved
26-39	Leaning refuted
11-25	Mostly refuted
0-10	Refuted

Insufficient-evidence gate

- Total adjusted evidence P + N is below the configured minimum for the claim type.
- No essential claim has adequate evidence.
- The only information comes from one interested source and cannot be independently checked.
- Key definitions, dates, or identities cannot be resolved.

10. Verdict confidence, context completeness, and final labels

Verdict Confidence

Component	Weight	Definition
Evidence coverage	30%	Weighted share of essential claims with adequate evidence.
Average evidence quality	25%	Weighted mean quality of accepted evidence.
Source independence	20%	Diversity of genuinely separate origins.
Evidence consistency	15%	Degree to which high-quality evidence points in one direction.
Primary/original-record access	10%	Availability and usability of original evidence.

Confidence base

$$0.30COV + 0.25QUAL + 0.20IND + 0.15CONS + 0.10PRI$$

Verdict Confidence

$$clamp(Confidence\ base - penalties, 0, 100)$$

Evidence consistency

$$CONS = 100 \times |P - N| / (P + N)$$

Equal high-quality support and contradiction produce low consistency; one dominant direction produces high consistency.

Typical unresolved problem	Default deduction
Essential term remains ambiguous	-15
Speaker or claimed date cannot be resolved	-10
Primary/original evidence is unavailable	-10
Major high-quality contradiction remains unresolved	-15
Evidence comes from one information or testimonial cluster	-15
Important source is inaccessible	-5
Translation is uncertain	-10
Edited media cannot be authenticated	-15
Actively developing event	-5 to -15

Confidence score	Label
85-100	Very high
70-84	High
50-69	Moderate
30-49	Low
0-29	Very low

Context Completeness

Context begins at 100 and loses points only when an omission materially changes how a reasonable reader would understand the claim or quotation.

General context failure	Deduction
Key term undefined or used inconsistently	-20
Important date range omitted	-15
Baseline or denominator omitted	-20
Relative change presented without absolute values	-15
Correlation described as causation	-25
Material qualifier omitted	-15
Comparison groups are not equivalent	-20
Unit or accounting measure changed	-20
Surrounding quote context changes the meaning	-25

Quote-specific context failure	Deduction
Conditional language such as "if," "could," or "might" removed	-25
Important preceding or following sentence omitted	-20
Time, location, or population scope omitted	-15
Speaker was quoting or describing another person	-25
Sarcasm, joke, or hypothetical presented literally	-25
Question presented as an assertion	-20
Translation removes a material qualifier	-20
Edited clip hides a correction or retraction	-25

Context Completeness
`clamp(100 - sum(material penalties), 0, 100)`

Misleading-but-technically-supported rule
When Evidence Support is at least 70 but Context Completeness is below 50, the default result is "Technically supported but misleading" rather than simply "Supported."

Final claim and attribution logic

Condition	Default output
Confidence below 35 or usable evidence too sparse	Insufficient evidence / unable to verify
Support 90-100 and context at least 70	Supported
Support at least 70 and context below 50	Technically supported but misleading
Support 75-89	Mostly supported
Support 60-74	Leaning supported
Support 40-59	Mixed or unresolved
Support 26-39	Leaning refuted
Support 11-25	Mostly refuted
Support 0-10	Refuted

Attribution condition	Default attribution label
Support at least 90, confidence at least 70, and high fidelity	Exact quote confirmed
Support at least 75, meaning preserved, wording differs	Accurate paraphrase
Support 55-74 or important wording disputed	Partially accurate / exact wording uncertain
Words confirmed but context below 50	Taken out of context
Support at most 25 with strong contrary attribution evidence	Misattributed or refuted
Confidence below 35 or evidence too sparse	Unable to verify

Combined result patterns

Attribution	Content	Combined report
Confirmed	Supported	The person said it, and the factual content is supported.
Confirmed	Refuted	The person said it, but the factual content is contradicted.
Misattributed	Supported	The underlying fact may be supported, but the quote is attributed to the wrong person.
Partial or out of context	Mixed	Some words are genuine, but the circulated version changes meaning and the factual claim remains disputed.
Unable to verify	Any content result	Attribution remains unresolved; factual content can still be evaluated separately.

11. He-said/she-said and disputed-attribution cases

The hardest attribution case occurs when one person says a statement was made, the alleged speaker denies it, and no recording or contemporaneous transcript exists. Elara.ai treats this as a disputed-attribution investigation, not a contest in which fame, confidence, or the number of repeated articles determines the result.

Testimony evaluation dimensions

Dimension	Questions
Direct access	Was the person present and able to hear or observe the statement?
Contemporaneity	Was the account recorded immediately or reconstructed much later?
Specificity	Does the account provide verifiable details rather than vague recollection?
Internal consistency	Has the account materially changed?
External corroboration	Do messages, calendars, notes, witnesses, or later actions independently support it?
Independence	Did witnesses reach their accounts independently or repeat one another?
Contradictory record	Is there a document, recording, or timeline that makes the account unlikely?
Interest or pressure	Does the source have a stake in the outcome? This lowers certainty but does not automatically make the account false.

Appropriate outcomes

Evidence pattern	Result
Authenticated recording or transcript matches the allegation	Attribution confirmed.
Authenticated record contradicts the alleged wording	Attribution refuted or misquoted.
Several independent contemporaneous witnesses plus records	Likely supported, with confidence tied to evidence quality.
One allegation, one denial, no corroboration	Unable to verify from available evidence.
Several reports all trace to one witness	One source cluster, not independent corroboration.
Accounts differ on minor wording but agree on meaning	Exact quotation uncertain; paraphrased meaning may be supported.
Accounts conflict on central meaning	Disputed and unresolved unless external evidence breaks the tie.

Required wording

When the evidence cannot resolve the dispute, state: "Available public evidence does not establish whether the statement was made." Do not label either person a liar, and do not treat a lack of evidence as proof that the event did not occur.

12. Article-level assessment and report presentation

Article-level dimensions

Metric	How it is calculated
Factual Accuracy	Importance-weighted average of atomic claim support scores, subject to essential-claim gates.
Verification Coverage	Weighted share of important claims that received adequate evidence.
Sourcing Transparency	Share of claims with identifiable sources, links, named authors, dates, methods, and visible corrections.
Source Independence	Independence of information origins behind major claims.
Context Completeness	Importance-weighted context score across major factual claims.
Headline Integrity	How accurately the headline represents the body and underlying evidence.
Numerical Accuracy	Accuracy of statistics, units, denominators, calculations, and time comparisons.
Attribution Accuracy	Accuracy and context of quotations and attributed statements.

Article Factual Accuracy

$$\text{sum}(S_j \times I_j) / \text{sum}(I_j)$$

S_j is the atomic claim support score; I_j is the importance weight 3, 2, or 1.

Article safeguards

- Unverified claims do not quietly count as accurate; they reduce verification coverage.
- A strongly refuted essential claim with high confidence caps the overall assessment.
- A misleading headline is scored separately from an otherwise accurate body.
- Predictions and opinions are labeled but excluded from factual-accuracy averages.
- The result describes the submitted article, not the publisher's permanent trustworthiness.

Completed report structure

Section	Contents
Overview	Verdict, score roles, summary, research timestamp, and limitations.
Attribution	Whether the statement was made; original record, speaker, date, venue, and wording match.
Context	Surrounding transcript, missing qualifiers, editing, translation, scope, and definitions.
Atomic claims	Individual support, confidence, importance, and unresolved gaps.
Evidence	Side-by-side supporting and contradicting passages.
Source graph	Original records, witnesses, derivative reports, and dependency clusters.
Timeline	When statements and records appeared and when corrections occurred.
Numerical audit	Formulas, source values, units, periods, denominators, and reproducible calculations.
Methodology	Scoring version, workflow version, model/prompt versions, search date, and calculation inputs.
Feedback and corrections	Controls for missed evidence, incorrect classifications, broken citations, appeals, and corrections.

Required product language

- Describe only the specific claim, article, or statement under review.

- Distinguish "not verified" from "false."
- Distinguish verified attribution from verified factual content.
- Clearly label allegations, testimony, predictions, opinions, and unresolved causation.
- Use exact dates and identify the evidence available as of the report date.
- Show the strongest credible contradiction rather than hiding disagreement.
- Allow users to open the source passage, table, page, or transcript timestamp for every conclusion.

Mandatory timestamp

Every report should state: "Evidence reviewed as of [date and time]. New evidence or corrections may change this assessment."

13. Storage, caching, updates, and reproducibility

Layer	Stored content
PostgreSQL	Sources, canonical URLs, metadata, claims, passages, extraction status, dependency edges, evidence links, calculations, scores, feedback, and versions.
pgvector	Passage embeddings and source-similarity vectors.
Object storage	Permitted snapshots, PDFs, user uploads, and generated exports.
Redis	Short-lived cache, distributed fetch locks, job state, rate limits, and progress events.

Snapshot and caching rules

- Store at minimum the content hash, retrieval time, parser version, and exact evidence excerpts included in the report.
- Reuse recently fetched content when the same canonical URL and revision are requested.
- Use different time-to-live values for stable filings, news pages, live blogs, and APIs.
- Use distributed locks so several workers do not fetch the same URL simultaneously.
- Refresh or invalidate when update headers, content hashes, or correction metadata change.
- A changed page creates a new snapshot rather than silently overwriting evidence used by an earlier report.

Update detection

`changed = new_content_hash != stored_content_hash`

Reproducibility rule

Every report records source versions, retrieval times, parser/extraction versions, model and prompt versions, workflow version, scoring methodology, queries, formulas, manual corrections, and reviewer overrides. A later webpage edit must not change what an earlier report claims it analyzed.

Core source records

```
Source
- id, canonical_url, domain, title, author, publisher
- published_at, updated_at, retrieved_at
- source_type, content_type, access_status
- content_hash, parser_name, parser_version
- snapshot_path, extraction_quality
- correction_status, robots_or_policy_status

Passage
- id, source_id, text, heading_path, page_or_position
- embedding, extraction_certainty

DependencyEdge
- parent_source_id, child_source_id
- relationship, confidence, detection_method
```


14. Failure handling, legal controls, privacy, and security

Failure handling and fallbacks

Failure	Allowed response
Timeout or temporary server error	Retry with bounded exponential backoff, then record failure.
Low-quality HTML extraction	Try a secondary extractor or Playwright when justified.
Rate limited	Respect retry timing, reduce concurrency, and continue with other sources.
Paywall or login	Do not bypass; use public alternatives or lawful user-provided text.
Unsupported format	Record the format and omit it from verified evidence.
Conflicting metadata	Preserve both values and lower extraction certainty.
No adequate sources	Return insufficient public evidence.
Search provider failure	Use a configured secondary provider or pause discovery while preserving current results.

Retry and research budget

- One or two network retries for transient failures.
- At most one expensive rendering fallback per important source.
- One broadened search round when evidence coverage remains below threshold.
- Total source and time budgets determined by Quick, Standard, or Deep mode.

Legal and ethical access rules

- Do not bypass authentication, paywalls, access controls, or anti-bot challenges.
- Respect applicable terms, rate limits, robots or site-access policies, and data-use restrictions.
- Store only the content required for verification and auditability.
- Prefer short evidence excerpts linked to the original source over republishing complete articles.
- Provide removal and correction procedures for stored snapshots where appropriate.

Privacy

- Do not crawl private profiles, private documents, or user accounts.
- Remove unnecessary personal data from logs and analytics.
- Apply retention limits to user uploads and source snapshots.
- Encrypt credentials and sensitive records in transit and at rest.

Security boundaries

Threat	Control
Server-side request forgery	URL allow rules, IP blocking, DNS revalidation, port and redirect checks.
Prompt injection	Treat retrieved text as untrusted data; separate content from agent instructions.
Malware or hostile files	Content-type allowlist, size limits, isolated parsing, and no execution.
Resource exhaustion	Per-user budgets, per-domain throttles, queue limits, response limits, and timeouts.
Credential leakage	Keep all provider credentials in server-side secret storage.
Cross-user data exposure	Authorization checks on every run, source, upload, snapshot, and report route.

15. Observability, evaluation, research modes, and governance

Operational metrics

Metric	Purpose
Search-to-fetch conversion	How many discovered results are useful enough to retrieve.
Extraction success rate	Share of fetched pages producing acceptable content.
Median fetch latency	User experience and capacity planning.
Playwright fallback rate	Whether browser rendering is being overused.
Cache hit rate	Effectiveness of reuse and cost control.
Duplicate-cluster rate	How much search-result volume is derivative reporting.
Evidence yield	Accepted evidence passages per fetched source.
Cost per verification	Search, model, storage, proxy, and compute expense.

Required quality evaluation

- Verdict accuracy and macro-F1 across supported, refuted, mixed, and insufficient-evidence classes.
- Attribution accuracy for quotations and alleged statements.
- Evidence precision and recall, including passage recall against human-required evidence.
- Citation entailment: each cited passage must actually support the sentence attached to it.
- URL relevance precision and primary-source recall.
- Extraction fidelity for text, numbers, dates, units, speakers, and quotations.
- Numerical accuracy and unit/date consistency.
- Source and duplicate-clustering accuracy.
- Confidence calibration error and Brier score.
- Unsupported-statement rate in generated reports.
- Latency and cost per verification.
- Security regression tests for SSRF, redirects, oversized files, hostile HTML, and prompt injection.

Research modes

Mode	Typical limits	Use
Quick	4-6 sources, one query round, no browser fallback unless essential	Fast headline or simple numerical check.
Standard	8-12 sources, support and contradiction paths, limited fallback	Default article or statement verification.
Deep	15-30 sources, broader provenance, multiple primary-source attempts	High-impact or complex investigation.

Governance requirements

- Calibrate thresholds on human-reviewed finance, science, technology, current-event, quote, and allegation cases before public launch.
- A confidence score of 80 should correspond to about 80 percent correctness over a sufficiently large benchmark; otherwise it is miscalibrated.
- Attach scoring formulas, model and prompt versions, workflow versions, retrieval timestamps, and source sets to every report.
- Provide appeals and corrections for users and affected speakers.
- Prevent automatic publication of high-impact allegations without stronger review controls.
- Re-test whenever models, search providers, weights, prompts, or extraction tools change.

System limitations shown to users

- Private, deleted, classified, paywalled, or inaccessible evidence may prevent verification.

- Anonymous sourcing and private conversations may be impossible to independently confirm.
- Breaking events and edited webpages can change after a report is generated.
- Primary sources, witnesses, transcripts, screenshots, and recordings can be mistaken, selective, manipulated, or deceptive.
- Automated extraction, OCR, translation, and transcription can misread tables, layouts, or meaning.
- Causal claims often remain uncertain even when the underlying events are well documented.
- High confidence is not certainty; it reflects evidence quality and coverage under a published methodology.

16. Worked examples

Example 1 - Article claim: "Company X doubled its profits last quarter"

The broad word "profits" is ambiguous. Elara.ai treats net income as the central ordinary interpretation while recording that operating income may have approximately doubled.

Evidence	Stance	Base quality q	Dependency m	Adjusted w
Audited filing: net income rose from \$1.20B to \$1.46B	-1.00	0.976	1.00	0.976
Company release: operating income approximately doubled	+0.50	0.800	0.35	0.280
Syndicated article repeating the release	+0.50	0.750	0.00	0.000

Support calculation

$P = (0.50 \times 0.280) + (0.50 \times 0.000) = 0.140; N = (1.00 \times 0.976) = 0.976$

Evidence Support

$100 \times 0.140 / (0.140 + 0.976) = 12.5 \rightarrow 13/100$ (Mostly refuted)

Confidence component	Score
Coverage	100
Average evidence quality	90
Source independence	55
Consistency	75
Primary-source access	100

Confidence

$0.30(100) + 0.25(90) + 0.20(55) + 0.15(75) + 0.10(100) = 84.75; \text{ minus 10 ambiguity penalty} = 74.75 \rightarrow 75/100$

Context Completeness is about 45/100 because "profits" is undefined and the wording switches between net income and operating income.

Final result

Mostly refuted. Net income rose by about 21.7 percent, not 100 percent. A different measure, operating income, approximately doubled. The wording is inaccurate under the ordinary net-income interpretation and incomplete about the accounting measure.

Retrieval handoff for the expanded causal claim

For "Company X doubled its profits last quarter because vehicle demand increased," the retrieval subsystem finds a regulatory filing, investor-relations results, a delivery release, an independent analyst report, and three syndicated copies. The copies are grouped into one derivative chain. The filing shows net income increased 21.7 percent, deliveries increased 8 percent, and the filing attributes the result to pricing, costs, and tax items; the causal claim remains unresolved.

Example 2 - Authentic quote with inaccurate or ambiguous content

Circulating statement: "The CEO said the company lost \$2 billion last year." An authenticated earnings-call transcript confirms the words. The audited filing reports a \$2.0 billion operating loss but a \$1.3 billion net loss.

Metric	Result
Attribution	Confirmed - 97/100
Quote Fidelity	95/100
Context Completeness	58/100
Factual Support	42/100 under the net-loss interpretation
Confidence	91/100

Final wording

The CEO made the statement. It is accurate only if "loss" refers to operating loss; the company's net loss was \$1.3 billion. The quote is authentic but financially ambiguous.

Example 3 - Real words, misleading edit

Viral version: "Unemployment will double." The recording contains the phrase, but the full sentence is: "Unemployment could double if the proposal fails." The edit removes both uncertainty and the condition.

Metric	Result
Attribution	Confirmed
Quote Fidelity	72/100
Context Completeness	18/100
Combined attribution label	Taken out of context

Example 4 - One allegation, one denial, no record

Person A says Person B made a private statement. Person B denies it. No recording, message, contemporaneous note, or independent witness is available. The accounts are direct but conflicting; neither independently corroborates itself, and duplicated reporting does not create additional origins.

Metric	Result
Attribution Support	Approximately neutral
Source Independence	Low
Verdict Confidence	Below 35
Attribution result	Unable to verify from available public evidence

17. Implementation sequence and appendices

Targeted retrieval implementation phases

Phase	Deliverables
Phase 1 - Minimum viable retrieval	Search integration with primary/support/counter groups; URL canonicalization and validation; static HTTP fetching; article and metadata extraction; page-aware PDF parsing; PostgreSQL source/passage tables; caching, timeouts, source limits, and explicit failure reasons.
Phase 2 - Quality and provenance	Embeddings and hybrid retrieval; content hashes and exact duplicates; near-duplicate clustering and citation edges; primary-source prioritization; domain diversity; correction and update detection.
Phase 3 - Production hardening	Isolated Playwright fallback; per-domain rate limits and compliance policies; object-storage snapshots and versioned evidence; monitoring and parser alerts; security regression tests; multiple search-provider abstraction and controlled failover.
Phase 4 - Advanced retrieval	Table-aware and chart-data extraction; OCR fallback; connectors for filings, legislation, courts, science, and statistics; multilingual query expansion and cross-language evidence matching; user-visible source timelines and snapshot comparison.

Default operational policies

Policy	Initial default
Allowed protocols	HTTP and HTTPS only
Redirect limit	3
Static request timeout	10-20 seconds total
Maximum normal response size	Configurable by type; reject unexpectedly large files
Playwright retries	Maximum one per important source
Search broadening	Maximum one additional round in Standard mode
Snapshot strategy	Evidence excerpt and hash always; full snapshot only when permitted
Derivative sources	Cluster before evidence independence is calculated
Inaccessible source	Record explicit status; never represent as verified

Default weight summary

Score	Default components
Evidence quality	Relevance 25%; Directness 20%; Claim-specific authority 20%; Transparency 15%; Temporal fit 10%; Extraction certainty 10%
Verdict Confidence	Coverage 30%; Average quality 25%; Independence 20%; Consistency 15%; Primary/original access 10%
Source Independence	Origin diversity 40%; Primary-source diversity 25%; Organizational diversity 20%; Method/dataset diversity 15%
Quote Fidelity	Wording 35%; Speaker 20%; Completeness 20%; Sequence 15%; Translation 10%

Final principle

Elara.ai should answer only what the evidence supports, showing what was found, how it was found, why each source was selected, source dependencies, inaccessible evidence, and how the published methodology produced the final label.